

Transcendental Algebraic Geometry

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October 22, 2025

Abstract

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1.1

Recall that the category of schemes and the category of analytic spaces are both full subcategories of locally ringed spaces.

Definition 1.1.1 (The Analytification of a Scheme)

Let X be a scheme locally of finite type over $\mathbb{C}$. Define the analytification of X to be the unique analytic space X^{an} together with a morphism of locally ringed space $X^{\text{an}} \rightarrow X$ satisfying the following universal property: For any analytic space Y , post composition with the morphism $X^{\text{an}} \rightarrow X$ gives a bijection of sets

$$\text{Hom}_{\mathbf{LRS}}(Y, X^{\text{an}}) = \text{Hom}_{\mathbf{An}}(Y, X^{\text{an}}) \cong \text{Hom}_{\mathbf{LRS}}(Y, X)$$

Recall that the Yoneda embedding $h : \mathbf{LRS} \rightarrow \mathbf{Set}^{\mathbf{LRS}}$ is the functor sending a locally ringed space X to the functor $h_X : \mathbf{LRS} \rightarrow \mathbf{Set}$ defined by $h_X(Y) = \text{Hom}_{\mathbf{LRS}}(Y, X)$. Then the universal property of the analytification means that the functor h_X restricted to analytic spaces $\mathbf{An}$ is representable.

One thinks of the morphism $X^{\text{an}} \rightarrow X$ as doing two things. It is an enlargement of the standard topology of X^{an} to the Zariski topology of X , and it introduces generic points outside of $X(\mathbb{C})$ (Indeed we will see that X^{an} has the same underlying set as $X(\mathbb{C})$).

Theorem 1.1.2

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the analytification of X exists and is unique up to unique isomorphism.

Definition 1.1.3 (The Analytification Functor)

Define the analytification functor

$$(-)^{\text{an}} : \mathbf{Sch}_{\mathbb{C}} \rightarrow \mathbf{An}$$

as follows.

- For each scheme X locally of finite type over $\mathbb{C}$, we have X^{an} is the analytification of X .
- For a morphism of schemes $f : X \rightarrow Y$ over $\mathbb{C}$, the universal property of Y^{an} induces a morphism $X \rightarrow Y^{\text{an}}$. Precomposing with the canonical morphism $X^{\text{an}} \rightarrow X$ gives the desired map $X^{\text{an}} \rightarrow Y^{\text{an}}$.

Proposition 1.1.4

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the underlying set of X^{an} is given by $X(\mathbb{C})$.